

Dipak Sumesara

Senior Backend Engineer

Node.js | TypeScript | Express.js | MongoDB | Microservices | AWS

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SUMMARY

Backend engineer with 8+ years in Node.js, TypeScript and MongoDB. Led the backend of a B2B operations platform for a US company from 2023 to 2026: scalable microservices on AWS, queue-based integrations, search and caching. Earlier built backends for a Singapore government health app, a fantasy sports platform with 100K+ concurrent users at peak match load, and an agritech supply-chain platform. Also builds generative AI features (LLMs, RAG, tool-calling agents) in TypeScript.

TECHNICAL SKILLS

- **Languages and frameworks:** TypeScript, JavaScript, Node.js, Express.js, NestJS, Fastify, React, Redux
- **APIs and auth:** REST APIs, GraphQL, WebSockets, Server-Sent Events, JWT, OAuth (Google), API keys
- **Databases and search:** MongoDB (NoSQL), PostgreSQL and MySQL (SQL), Redis, Elasticsearch, schema design, data modeling
- **Architecture:** Microservices, distributed systems, event-driven architecture, API Gateway, message queues (SQS, BullMQ), pub/sub, system design
- **Cloud and DevOps:** AWS (Lambda, API Gateway, SQS, S3, CloudFront), Docker, Git, GitHub Actions (CI/CD), Sentry
- **Testing:** Jest, Vitest, unit and integration testing
- **AI and LLMs:** Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), generative AI, embeddings and vector search, tool calling (function calling), agent orchestration, prompt engineering, Zod-validated outputs, Anthropic Claude API, OpenAI API, Ollama (Llama, Qwen)
- **AI coding tools:** Claude Code, Cursor, GitHub Copilot, OpenAI Codex

PROFESSIONAL EXPERIENCE

SwagUp

Mar 2023 - Apr 2026

Senior Backend Engineer | Remote

Tech stack: Node.js, TypeScript, Express.js, MongoDB, Redis, Elasticsearch, GraphQL, AWS (Lambda, SQS, S3, CloudFront), Docker, GitHub Actions, React, Redux, React Query, MUI

Aloha, SwagUp's B2B operations platform (vendor, distributor and admin workflows), and the customer-facing swagup.com. Joined as a Full Stack Developer, moved to the backend as Senior Backend Engineer, and later led the Aloha backend.

- Cut API response times from seconds to milliseconds with Redis caching, query optimization and database indexing.
- Led system design and delivery for the Aloha backend alongside US product and engineering teams, and set the team's Jest testing and code-review standards.
- Split the platform into microservices behind an API Gateway on AWS Lambda, with a central service for authentication (JWT, API keys, Google OAuth), authorization and sessions.
- Built a Node.js-to-Salesforce data sync on SQS with a dead-letter queue to capture and recover failed messages. Ran other async jobs and event-driven flows on pub/sub, SQS and BullMQ, with retries, monitoring and observability patterns.
- Built product search on Elasticsearch (~20K products), with indexing pipelines that kept the index consistent with the source data.
- Kept the platform available with connection pooling, read/write splitting, replication, failover and load balancing. Handled production incidents with Sentry, logging and alerting, including root cause analysis, rollbacks and hotfixes.
- Designed schemas and data models, versioned the APIs and enforced API contracts through specs, schema enforcement and response transformers.
- Worked with DevOps on GitHub Actions CI/CD and release workflows, containerized services with Docker for Kubernetes deployments, and ran off-hours production releases under high-availability SLAs.
- Built the Aloha admin, quote and pricing screens in React, using Redux, React Query and MUI.
- On swagup.com: built REST and GraphQL APIs for order tracking and a multi-tier pricing engine with real-time quotes and dynamic discounts. Stored product media in S3 and served images through CloudFront.

CoffeeBeans Consulting

Jun 2021 - Mar 2023

Software Engineer / Backend Lead | Remote

Tech stack: Node.js, Express.js, TypeScript, MongoDB, PostgreSQL, Strapi, AWS (S3, CloudFront)

- Built Node.js backend services for Healthy 365, the Singapore Health Promotion Board's citizen health app, in a 10-person cross-functional team.
- Shipped under enforced quality gates: 90% minimum unit-test coverage with commits blocked below it, strict ESLint rules, pair programming, dual QA and staged Dev, QA, UAT and Production releases.
- Owned the backend architecture of Salam Kisan, an agricultural supply-chain platform, on Strapi CMS: REST APIs, an Open edX integration for learning content and automated data-sync workflows.

- Led the 3-engineer Salam Kisan backend team across sprint delivery, code quality and production support.

Yudiz Solutions

May 2020 - May 2021

Software Engineer | Ahmedabad, India

Tech stack: Node.js, Express.js, TypeScript, MongoDB, Redis, WebSockets, React, Redux, AWS (S3, CloudFront)

- Built real-time backend features for 11Wickets, a fantasy sports platform (Dream11 competitor) serving 100K+ concurrent users at peak match load, using WebSockets for live score updates and Redis caching for high-concurrency reads.
- Built WebSocket live chat for use during matches, and mentored junior developers.

Biztech Consulting & Solutions

Jan 2019 - Mar 2020

Software Engineer | Ahmedabad, India

Tech stack: Angular, TypeScript, MySQL

- Proposed and led the v2.0 rewrite of BrushYourIdeas, moving the product from a legacy Angular codebase to the latest Angular with a modular architecture, clean design patterns and secure API integrations. The founding team adopted it.
- Rebuilt the product as independent feature modules. Customers chose a Professional, Ultimate or Custom pack, and each pack was assembled by picking the matching module folders, replacing the earlier all-or-nothing model.

Angular Minds

Dec 2017 - Sep 2018

Software Engineer (Intern to Full-Time) | Pune, India

Tech stack: Node.js, NestJS, Express.js, TypeScript, MongoDB, Angular, Ionic

- Built Numnu, a food-event discovery app, with a NestJS and MongoDB backend and an Angular/Ionic frontend. It surfaced events by proximity and date range and had an event-guide mode used at the Royal Manitoba Winter Fair.

PROJECTS

AI Integration Discovery Agent

Sep 2026

github.com/dipak-sumesara/ai-integration-discovery-agent

Tech stack: Node.js, TypeScript, Express.js, Anthropic Claude API, Ollama, Zod, Vitest

- Built a proof-of-concept agent that reads API documentation from a Markdown file, probes a mock API through tool calls (authentication, users, groups, roles, scopes) and produces a Zod-validated connector specification.
- Wrote the multi-step agent loop without a framework, with swappable LLM providers (Anthropic Claude, Ollama).
- Tested the harness with mocked providers and covered the HTTP tool's SSRF protections.

AI Commerce Assistant

Sep 2026

github.com/dipak-sumesara/ai-commerce-assistant

Tech stack: Node.js, TypeScript, Fastify, React, Vite, Ollama (Qwen, nomic-embed-text), Zod, Vitest

- Turned natural-language shopping queries into Zod-validated structured filters with Ollama, then applied filtering and ranking in deterministic TypeScript, behind a guardrail that refuses unrelated requests before retrieval or generation.
- Built a lightweight RAG pipeline that chunks Markdown knowledge documents, embeds them through Ollama and retrieves by cosine similarity in memory to ground answers.
- Wrote tests that mock the Ollama layer, so the suite runs without a model.

Reverie: AI Character Chat

May 2026

github.com/dipak-sumesara/ai-chat-app

Tech stack: Node.js, TypeScript, Express.js, MongoDB, Redis, Ollama (Llama), React, Vite, Tailwind CSS, Zustand

- Built conversation memory that keeps long chats inside the model's context window and cuts prompt token usage: a recent-message window plus AI-generated summaries cached in Redis and refreshed after each exchange.
- Streamed replies token by token over Server-Sent Events (SSE) and stored editable, branching chat history in MongoDB.

EDUCATION

Master of Computer Applications (MCA) | Savitribai Phule Pune University, Pune

2018

Bachelor of Computer Applications (BCA) | Krantiguru Shyamji Krishna Verma Kachchh University, Bhuj

2015